What is the purpose of the ARP protocol?

To map a known IP address to a physical MAC address so that devices can communicate over a local network.

What problem does ARP solve in a network?

Computers use IP addresses to identify each other, but local hardware (Ethernet/Wi-Fi) requires a MAC address to deliver data frames. ARP finds the hardware address associated with a specific IP.

At which OSI layer does ARP operate?

It operates at Data Link Layer, though it functions as a bridge to the Network Layer.

What is the difference between an IP address and a MAC address?: An IP address is a logical identifier that is assigned to a device based on a network, while a MAC address is a permanent, unique physical identifier burned into the hardware.

Filter only ARP packets. What display filter did you use?: `arp`

```
PS C:\Program Files\Wireshark> .\tshark.exe -r "pracnet.net-arp.pcapng" -Y "arp"
    1   0.000000 00:53:ff:ff:aa:aa → Broadcast    ARP 60 Who has 10.0.0.22? Tell 10.0.0.11
    2   0.000343 00:53:ff:ff:bb:bb → 00:53:ff:ff:aa:aa ARP 60 10.0.0.22 is at 00:53:ff:ff:bb:bb
```

Identify an ARP Request packet: What is the source IP?: 10.0.0.11

Identify an ARP Request packet: What is the destination IP?: 10.0.0.22

Identify an ARP Request packet: What MAC address is being requested?:

Identify the corresponding ARP Reply: What MAC address is returned?: 00:53:ff:ff:bb:bb

Identify the corresponding ARP Reply: Which device owns that IP?: 10.0.0.2

What is the difference in structure between ARP Request and ARP Reply?

APR Request: Destination MAC, Operation, Target MAC, Target IP.

APR Reply: Destination MAC, Operation, Sender MAC, Target MAC

Why is the destination MAC address in an ARP request often ff:ff:ff:ff:ff:ff?: This is the Broadcast Address. Since the sender doesn't know which device has the IP, it sends the request to every device on the local network segment.

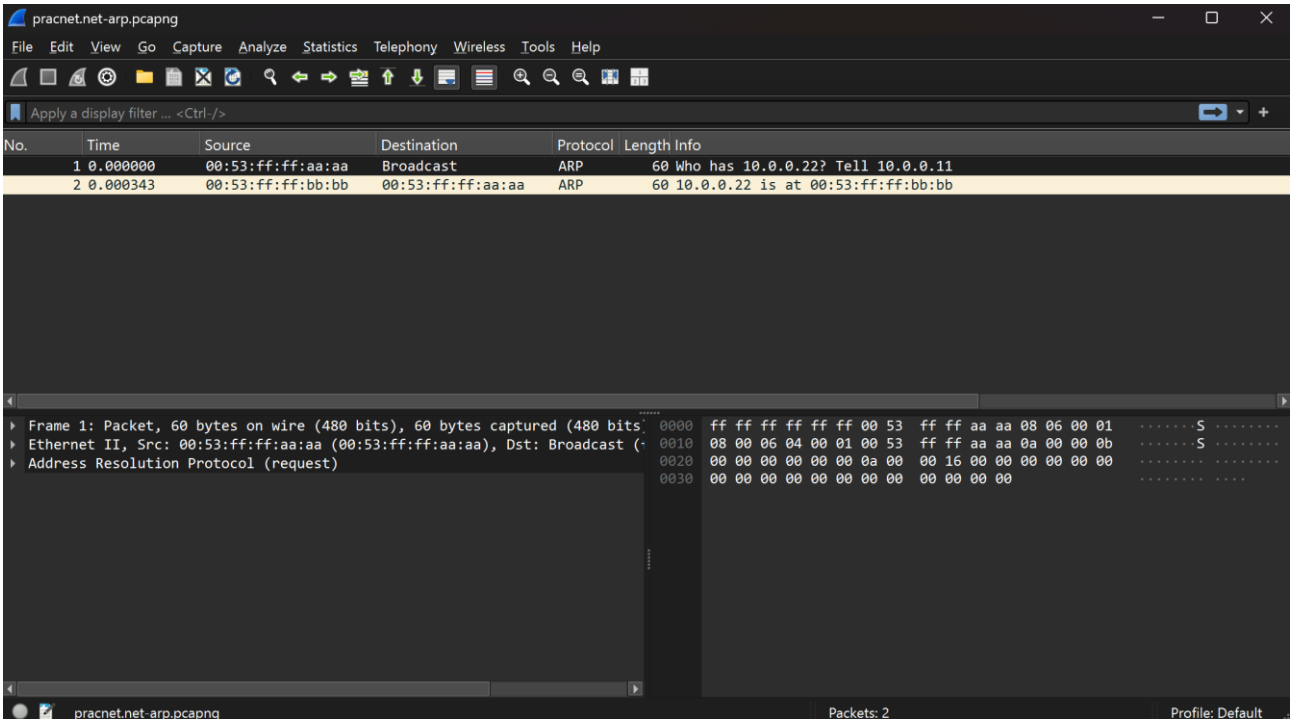
What does the phrase "Who has IP X.X.X.X?" mean in ARP?: It is a broadcast inquiry asking the device assigned that specific IP address to provide its physical MAC address.

How does a device respond to an ARP request?: If the Target IP in the request matches the device's own IP, it generates an ARP Reply containing its MAC address and sends it directly to the requester.

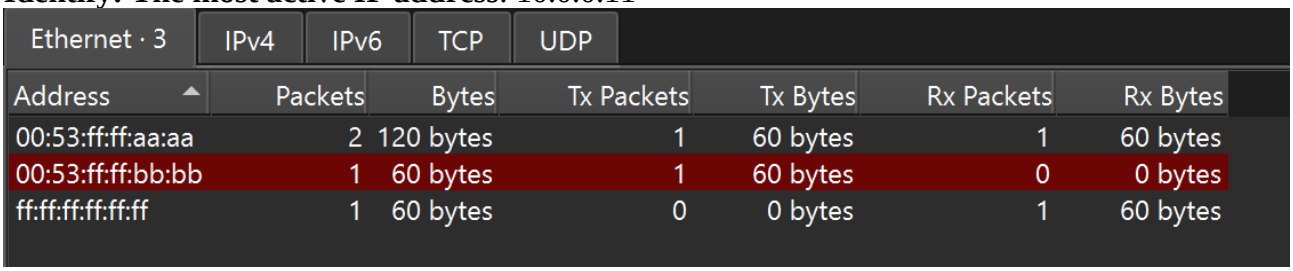
What is an ARP cache?: A temporary table stored in a device's RAM that maps IP addresses to MAC addresses so the device doesn't have to send a request for every single packet.

How long are entries typically stored?: Usually between 2 and 20 minutes, depending on the operating system's configuration.

How many ARP packets are in the capture file?: *2*

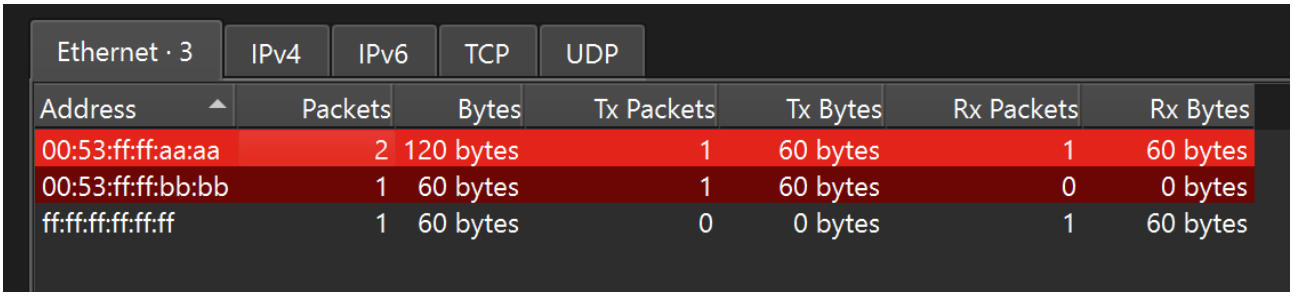


Identify: The most active IP address: 10.0.0.11



Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
00:53:ff:ff:aa:aa	2	120 bytes	1	60 bytes	1	60 bytes
00:53:ff:ff:bb:bb	1	60 bytes	1	60 bytes	0	0 bytes
ff:ff:ff:ff:ff:ff	1	60 bytes	0	0 bytes	1	60 bytes

Identify: The most active MAC address: 10.0.0.11



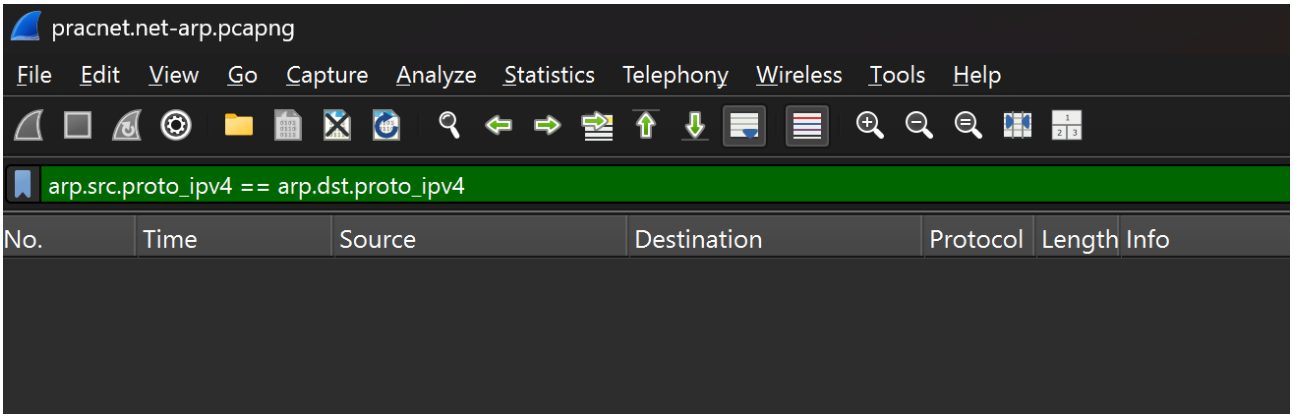
Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
00:53:ff:ff:aa:aa	2	120 bytes	1	60 bytes	1	60 bytes
00:53:ff:ff:bb:bb	1	60 bytes	1	60 bytes	0	0 bytes
ff:ff:ff:ff:ff:ff	1	60 bytes	0	0 bytes	1	60 bytes

Are there repeated ARP requests for the same IP? Why might this happen?: No

Reasons why might it happen:

- **The Target is Offline:** The device being looked for is powered off, disconnected from the cable, or has its Wi-Fi turned off.
- **Firewall Blocking:** The target device is online but configured to ignore or drop ARP requests for security reasons.
- **Packet Loss:** The network is congested or has a faulty switch/cable, causing the request or the reply to be dropped before it reaches its destination.
- **Wrong Subnet:** The requester is looking for an IP address that doesn't exist on that specific local network segment.

Do you observe any gratuitous ARP packets? What is their purpose? Their purpose is to announce a new IP/MAC to the network or to detect IP address conflicts.



Why is ARP considered insecure?:

It is a stateless and unauthenticated protocol. Devices accept ARP replies even if they never sent a request, and there is no way to verify if the sender is who they claim to be.

Does ARP include any authentication mechanism?: No.

What could happen if a malicious device sends fake ARP replies?

It can redirect traffic intended for a legitimate device (like the router) to itself, leading to data theft or service disruption.

Explain the concept of ARP spoofing (ARP poisoning)

An attacker sends falsified ARP messages onto a local network to link their MAC address with the IP address of a legitimate server or default gateway.

How could an attacker use ARP spoofing to intercept traffic?

By poisoning the ARP cache of a victim, the attacker ensures all traffic meant for the gateway is sent to the attacker's machine first.

What is a Man-in-the-Middle (MITM) attack in the context of ARP?

An attack where the perpetrator poisons both the victim and the gateway, sitting between them to capture, read, or modify all passing data.

Why is ARP spoofing effective in local networks but not across the internet?

ARP is a Layer 2 protocol that does not cross routers, it only functions within a single local broadcast domain (LAN).

What are potential signs of ARP-based attacks in packet captures?: Multiple IP addresses associated with a single MAC address, a high frequency of unsolicited ARP replies, or sudden changes in the MAC address of the default gateway.